

Week 4 Practical Laboratory

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Week 4 Practical Laboratory

Linux Security Monitoring and Auditing

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Executive Summary

A Linux security monitoring and auditing assessment was conducted to evaluate audit logging, system log monitoring, security event analysis, and system hardening controls. The assessment utilized auditd, journalctl, authentication logs, and the Lynis security auditing tool.

The review established that monitoring and auditing capabilities were available within the test environment, although some audit subsystem functionality was restricted by environment limitations. Log analysis identified authentication activity, privileged command execution, and system events requiring monitoring. The Lynis assessment further identified security hardening opportunities that could improve the overall security posture of the Linux host.

The three most significant findings were:

1. Authentication failures requiring monitoring and investigation.
2. Limited audit visibility due to audit subsystem restrictions.
3. Security hardening recommendations identified by Lynis.

Recommendations include strengthening monitoring processes, improving audit coverage, implementing Lynis recommendations, and integrating Linux logs into an enterprise SIEM platform.

1. Scope and Authorisation

The assessment was performed within the assigned Ubuntu Linux laboratory environment. The scope included:

- Auditd configuration and audit rule management.
- Linux system log analysis.
- Authentication and privilege-use monitoring.
- Security hardening review using Lynis.
- Governance and continuous monitoring mapping.

Testing activities were limited to the assigned laboratory environment and authorized educational exercises.

2. Methodology

The following assessment activities were performed:

Auditd Review

- Installation and verification of auditd.
- Configuration of custom audit monitoring rules.
- Review of audit rule loading and event reporting.

Log Analysis

- System log review using journalctl.
- Authentication log review.
- Privilege escalation monitoring.
- Error and warning analysis.

Security Hardening Assessment

- Execution of Lynis system audit.
- Review of security findings.
- Evaluation of hardening recommendations.

Governance Mapping

- Mapping technical findings to control monitoring.
- SIEM integration analysis.
- Escalation and remediation planning.

Evidence Bundle 1 – Auditd Configuration and Events

Objective

To configure and validate Linux auditing controls capable of monitoring sensitive files and security-related activities.

Activities Performed

Configured custom audit monitoring rules for:

- /etc/passwd
- /etc/shadow
- Program execution monitoring
- Authentication log monitoring

Audit Rules Configured

```
-w /etc/passwd -p rwx -k passwd_changes
```

```
-w /etc/shadow -p rwx -k shadow_changes
```

```
-a always,exit -F arch=b64 -S execve -k program_execution
```

```
-a always,exit -F arch=b32 -S execve -k program_execution
```

```
-w /var/log/auth.log -p wa -k auth_failures
```

Observations

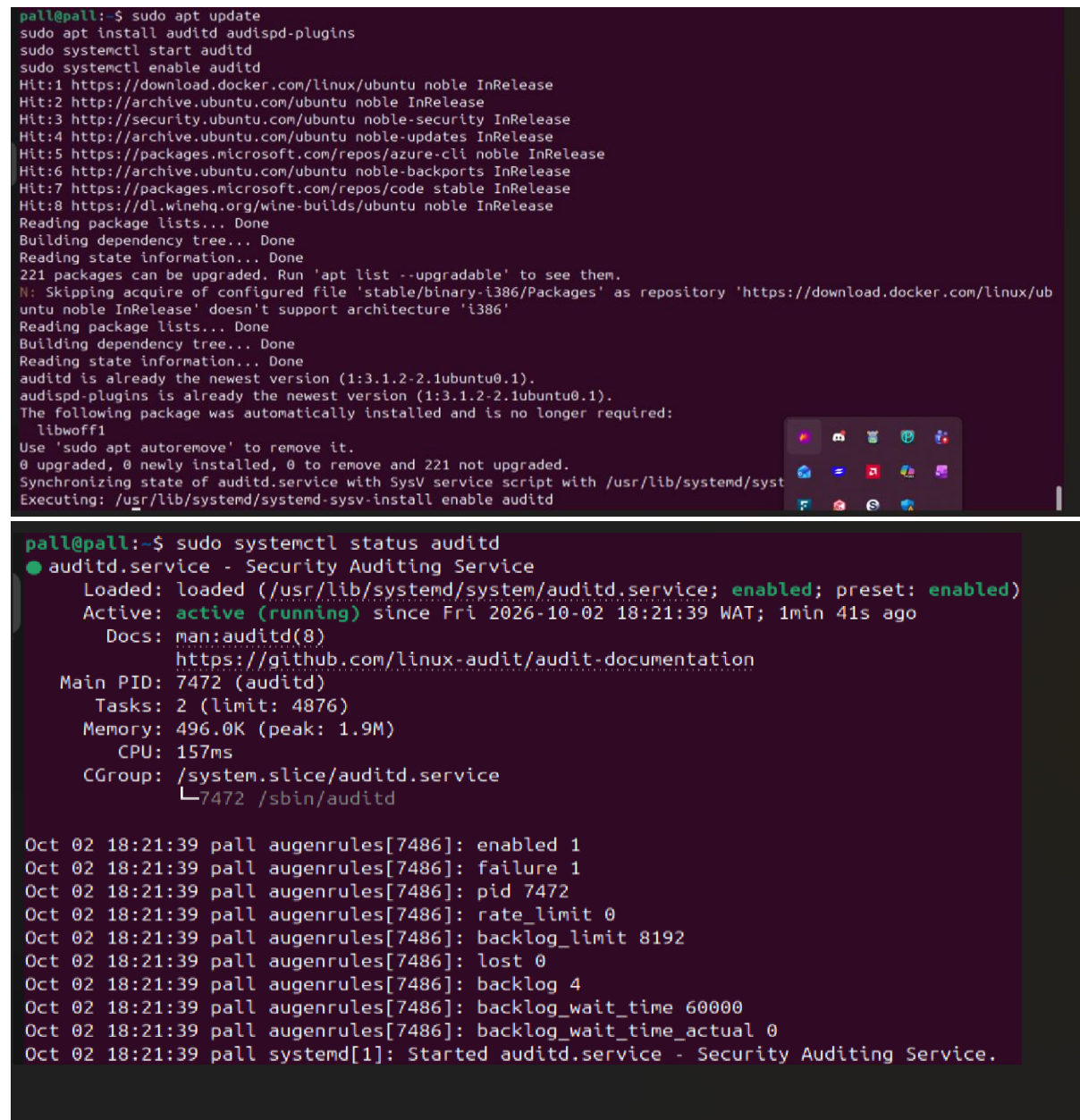
The auditctl utility was successfully installed and custom rules were created.

The laboratory environment restricted access to certain kernel-level audit functions and generated the following message:

Error sending rule list data request (Operation not permitted)

This indicates that audit rule verification was partially restricted by the underlying environment.

Screenshot 1: auditd installation

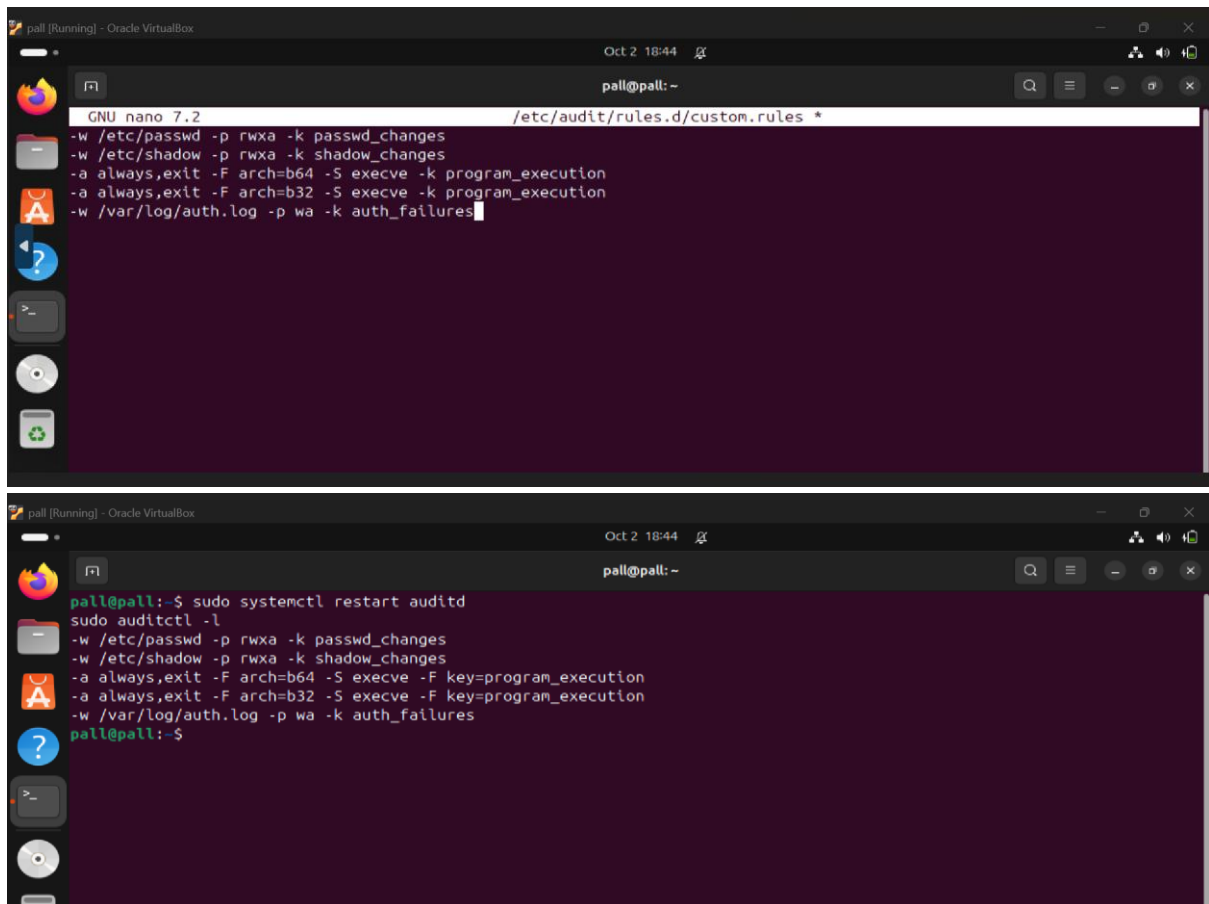


```
pall@pall:~$ sudo apt update
sudo apt install auditd audispd-plugins
sudo systemctl start auditd
sudo systemctl enable auditd
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 https://packages.microsoft.com/repos/azure-cli noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:7 https://packages.microsoft.com/repos/code stable InRelease
Hit:8 https://dl.winehq.org/wine-builds/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
221 packages can be upgraded. Run 'apt list --upgradable' to see them.
N: Skipping acquire of configured file 'stable/binary-i386/Packages' as repository 'https://download.docker.com/linux/ubuntu noble InRelease' doesn't support architecture 'i386'
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.1ubuntu0.1).
audispd-plugins is already the newest version (1:3.1.2-2.1ubuntu0.1).
The following package was automatically installed and is no longer required:
  libwoff1
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 221 not upgraded.
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install enable auditd

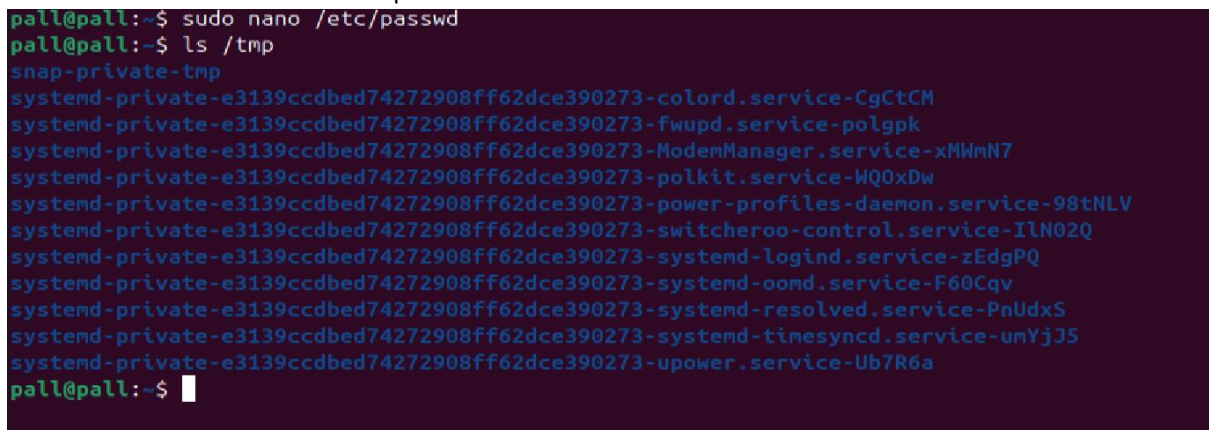
pall@pall:~$ sudo systemctl status auditd
● auditd.service - Security Auditing Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-10-02 18:21:39 WAT; 1min 41s ago
     Docs: man:auditd(8)
           https://github.com/linux-audit/audit-documentation
   Main PID: 7472 (auditd)
    Tasks: 2 (limit: 4876)
   Memory: 496.0K (peak: 1.9M)
      CPU: 157ms
   CGroup: /system.slice/auditd.service
           └─7472 /sbin/auditd

Oct 02 18:21:39 pall augenrules[7486]: enabled 1
Oct 02 18:21:39 pall augenrules[7486]: failure 1
Oct 02 18:21:39 pall augenrules[7486]: pid 7472
Oct 02 18:21:39 pall augenrules[7486]: rate_limit 0
Oct 02 18:21:39 pall augenrules[7486]: backlog_limit 8192
Oct 02 18:21:39 pall augenrules[7486]: lost 0
Oct 02 18:21:39 pall augenrules[7486]: backlog 4
Oct 02 18:21:39 pall augenrules[7486]: backlog_wait_time 60000
Oct 02 18:21:39 pall augenrules[7486]: backlog_wait_time_actual 0
Oct 02 18:21:39 pall systemd[1]: Started auditd.service - Security Auditing Service.
```

Screenshot 2: custom.rules configuration



Screenshot 3: auditctl output



Screenshot 4: ausearch/aureport evidence

```
pall@pall:~$ sudo ausearch -k passwd_changes
----
time->Fri Oct 2 18:44:51 2026
type=PROCTITLE msg=audit(1790963091.816:384): proctitle=2F7362696E2F617564697463746C002D52002F6574632F61756469742F617564
69742E72756C6573
type=SYSCALL msg=audit(1790963091.816:384): arch=c000003e syscall=44 success=yes exit=1084 a0=3 a1=7ffd6a4c1210 a2=43c a
3=0 items=0 ppid=8716 pid=8730 auid=4294967295 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=(none) ses=42
94967295 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1790963091.816:384): auid=4294967295 ses=4294967295 subj=unconfined op=add_rule key="passwd
_changes" list=4 res=1
----
time->Fri Oct 2 18:44:51 2026
type=PROCTITLE msg=audit(1790963091.836:393): proctitle=7375646F00617564697463746C002D6C
type=PATH msg=audit(1790963091.836:393): item=0 name="/etc/passwd" inode=796256 dev=08:02 mode=0100644 ouid=0 ogid=0 rde
v=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790963091.836:393): cwd="/home/pall"
type=SYSCALL msg=audit(1790963091.836:393): arch=c000003e syscall=257 success=yes exit=3 a0=ffffff9c a1=73d5b23d0339 a2=
80000 a3=0 items=1 ppid=5195 pid=8738 auid=1000 uid=1000 gid=1000 euid=0 suid=0 fsuid=0 egid=1000 sgid=1000 fsgid=1000 t
ty=pts0 ses=6 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_changes"
----
time->Fri Oct 2 18:44:51 2026
type=PROCTITLE msg=audit(1790963091.836:394): proctitle=7375646F00617564697463746C002D6C
type=PATH msg=audit(1790963091.836:394): item=0 name="/etc/passwd" inode=796256 dev=08:02 mode=0100644 ouid=0 ogid=0 rde
v=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790963091.836:394): cwd="/home/pall"
type=SYSCALL msg=audit(1790963091.836:394): arch=c000003e syscall=257 success=yes exit=5 a0=ffffff9c a1=73d5b23d0339 a2=
80000 a3=0 items=1 ppid=5195 pid=8738 auid=1000 uid=1000 gid=1000 euid=0 suid=0 fsuid=0 egid=1000 sgid=1000 fsgid=1000 t
ty=pts0 ses=6 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_changes"
----
pall@pall:~$ sudo ausearch -k program_execution
----
time->Fri Oct 2 18:44:51 2026
type=PROCTITLE msg=audit(1790963091.819:386): proctitle=2F7362696E2F617564697463746C002D52002F6574632F61756469742F617564
69742E72756C6573
type=SOCKADDR msg=audit(1790963091.819:386): saddr=10000000000000000000000000000000
type=SYSCALL msg=audit(1790963091.819:386): arch=c000003e syscall=44 success=yes exit=1076 a0=3 a1=7ffd6a4c1210 a2=434 a
3=0 items=0 ppid=8716 pid=8730 auid=4294967295 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=(none) ses=42
94967295 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1790963091.819:386): auid=4294967295 ses=4294967295 subj=unconfined op=add_rule key="progra
m_execution" list=4 res=1
----
time->Fri Oct 2 18:44:51 2026
type=PROCTITLE msg=audit(1790963091.819:387): proctitle=2F7362696E2F617564697463746C002D52002F6574632F61756469742F617564
69742E72756C6573
type=SOCKADDR msg=audit(1790963091.819:387): saddr=10000000000000000000000000000000
type=SYSCALL msg=audit(1790963091.819:387): arch=c000003e syscall=44 success=yes exit=1076 a0=3 a1=7ffd6a4c1210 a2=434 a
3=0 items=0 ppid=8716 pid=8730 auid=4294967295 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=(none) ses=42
94967295 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1790963091.819:387): auid=4294967295 ses=4294967295 subj=unconfined op=add_rule key="progra
m_execution" list=4 res=1
----
time->Fri Oct 2 18:44:51 2026
type=PROCTITLE msg=audit(1790963091.836:392): proctitle=7375646F00617564697463746C002D6C
type=PATH msg=audit(1790963091.836:392): item=1 name="/lib64/ld-linux-x86-64.so.2" inode=148616 dev=08:02 mode=0100755 o
uid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1790963091.836:392): item=0 name="/usr/bin/sudo" inode=132452 dev=08:
dev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790963091.836:392): cwd="/home/pall"
----
pall@pall:~$ sudo ausearch -k auth_failures
----
time->Fri Oct 2 18:44:51 2026
type=PROCTITLE msg=audit(1790963091.821:388): proctitle=2F7362696E2F617564697463746C002D52002F6574632F61756469742F617564
69742E72756C6573
type=PATH msg=audit(1790963091.821:388): item=0 name="/var/log/" inode=1048585 dev=08:02 mode=040775 ouid=0 ogid=102 rde
v=00:00 nametype=PARENT cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1790963091.821:388): cwd="/"
type=SOCKADDR msg=audit(1790963091.821:388): saddr=10000000000000000000000000000000
type=SYSCALL msg=audit(1790963091.821:388): arch=c000003e syscall=44 success=yes exit=1088 a0=3 a1=7ffd6a4c1210 a2=440 a
3=0 items=1 ppid=8716 pid=8730 auid=4294967295 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=(none) ses=42
94967295 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1790963091.821:388): auid=4294967295 ses=4294967295 subj=unconfined op=add_rule key="auth_f
ailures" list=4 res=1
pall@pall:~$
```

```
pall@pall:~$ sudo aureport
```

Summary Report

=====

```
Range of time in logs: 10/02/2026 18:21:39.552 - 10/02/2026 18:52:46.628
Selected time for report: 10/02/2026 18:21:39 - 10/02/2026 18:52:46.628
Number of changes in configuration: 11
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 1
Number of failed authentications: 0
Number of users: 3
Number of terminals: 7
Number of host names: 1
Number of executables: 14
Number of commands: 15
Number of files: 17
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 2
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 4
Number of process IDs: 42
Number of events: 374
```

```
pall@pall:~$ sudo aureport --failed
```

Failed Summary Report

=====

```
Range of time in logs: 10/02/2026 18:21:39.552 - 10/02/2026 18:53:18.137
Selected time for report: 10/02/2026 18:21:39 - 10/02/2026 18:53:18.137
Number of changes in configuration: 0
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 0
Number of users: 1
Number of terminals: 1
Number of host names: 0
Number of executables: 1
Number of commands: 1
Number of files: 1
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 2
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 1
Number of process IDs: 1
Number of events: 2
```

```
pall@pall:~$ sudo aureport --login

Login Report
=====
# date time auid host term exe success event
=====
<no events of interest were found>
```

Interpretation

Audit monitoring rules were configured successfully. Full validation of audit events was partially constrained by environment limitations affecting audit subsystem access.

Evidence Bundle 2 - Linux Log Analysis

journalctl Review

The system journal was reviewed to identify operational and security-relevant events.

Commands Executed

```
journalctl --since today
```

```
journalctl -p err
```

Observations

- System events were successfully recorded.
- Error-level messages were available for review.
- Logging functionality was operational.

Authentication and Privilege Review

Commands executed:

```
grep -i sudo /var/log/auth.log
```

```
grep -i "failed password" /var/log/auth.log
```

Event Analysis Table

Event	Source	Security Significance	Action
Failed authentication attempts	auth.log	Possible unauthorized access attempts	Monitor and investigate repeated failures
Sudo command execution	auth.log	Privileged activity detected	Review privileged user actions
System warnings/errors	journalctl/syslog	Potential service or configuration issues	Investigate recurring entries

Screenshots

Screenshot 5: journalctl output

```
pall@pall:~$ sudo journalctl
Jul 16 14:54:09 pall systemd[3844]: Queued start job for default target default.target.
Jul 16 14:54:09 pall systemd[3844]: Created slice app.slice - User Application Slice.
Jul 16 14:54:09 pall systemd[3844]: Created slice session.slice - User Core Session Slice.
Jul 16 14:54:09 pall systemd[3844]: Started ubuntu-report.path - Pending report trigger for Ubuntu Report.
Jul 16 14:54:09 pall systemd[3844]: Started launchpadlib-cache-clean.timer - Clean up old files in the Launchpadlib cache.
Jul 16 14:54:09 pall systemd[3844]: Started snap.firmware-updater.firmware-notifier.timer - Timer firmware-notifier for snap.
Jul 16 14:54:09 pall systemd[3844]: Reached target paths.target - Paths.
Jul 16 14:54:09 pall systemd[3844]: Reached target timers.target - Timers.
Jul 16 14:54:09 pall systemd[3844]: Starting dbus.socket - D-Bus User Message Bus Socket...
Jul 16 14:54:09 pall systemd[3844]: Listening on dirnmgr.socket - GnuPG network certificate management daemon.
Jul 16 14:54:09 pall systemd[3844]: Listening on gcr-ssh-agent.socket - GCR ssh-agent wrapper.
Jul 16 14:54:09 pall systemd[3844]: Listening on gnome-keyring-daemon.socket - GNOME Keyring daemon.
Jul 16 14:54:09 pall systemd[3844]: Listening on gpg-agent-browser.socket - GnuPG cryptographic agent and passphrase cache.
Jul 16 14:54:09 pall systemd[3844]: Listening on gpg-agent-extra.socket - GnuPG cryptographic agent and passphrase cache.
Jul 16 14:54:09 pall systemd[3844]: Starting gpg-agent-ssh.socket - GnuPG cryptographic agent (ssh-agent emulation)...
Jul 16 14:54:09 pall systemd[3844]: Listening on gpg-agent.socket - GnuPG cryptographic agent and passphrase cache.
Jul 16 14:54:09 pall systemd[3844]: Listening on keyboxd.socket - GnuPG public key management service.
Jul 16 14:54:09 pall systemd[3844]: Listening on pipewire-pulse.socket - PipeWire PulseAudio.
Jul 16 14:54:09 pall systemd[3844]: Listening on pipewire.socket - PipeWire Multimedia System Sockets.
Jul 16 14:54:09 pall systemd[3844]: Listening on pk-debconf-helper.socket - debconf communication socket.
Jul 16 14:54:09 pall systemd[3844]: Listening on snapd.session-agent.socket - REST API socket for snapd user session agent.
Jul 16 14:54:09 pall systemd[3844]: Listening on speech-dispatcher.socket - Speech Dispatcher Socket.
Jul 16 14:54:09 pall systemd[3844]: Listening on dbus.socket - D-Bus User Message Bus Socket.
Jul 16 14:54:09 pall systemd[3844]: Listening on gpg-agent-ssh.socket - GnuPG cryptographic agent (ssh-agent emulation).
Jul 16 14:54:09 pall systemd[3844]: Reached target sockets.target - Sockets.
Jul 16 14:54:09 pall systemd[3844]: Reached target basic.target - Basic System.
Jul 16 14:54:09 pall systemd[3844]: gnome-initial-setup-copy-worker.service - GNOME Initial Setup Copy Worker Service.
Jul 16 14:54:09 pall systemd[3844]: Started pipewire.service - PipeWire Multimedia Service.
Jul 16 14:54:09 pall systemd[3844]: Started filter-chain.service - PipeWire filter chain.
Jul 16 14:54:09 pall systemd[3844]: Started snap.snapd-desktop-integration.snapd-desktop-integration.service - Snapd Desktop Integration Service.
```

```
pall@pall:~$ sudo journalctl -u ssh
-- No entries --
```

```
pall@pall:~$ sudo journalctl -p err
Jul 16 14:54:10 pall systemd[3844]: Failed to start app-gnome-xdg\x2dduser\x2ddirs-4100.scope - Application launched by user.
Jul 16 15:41:06 pall sudo[5958]: pam_unix(sudo:auth): conversation failed
Jul 16 15:41:06 pall sudo[5958]: pam_unix(sudo:auth): auth could not identify password for [pall]
Jul 16 18:24:05 pall systemd[1]: snapd.service: Watchdog timeout (limit 5min)!
Jul 16 17:14:10 pall systemd[1]: systemd-udevd.service: Watchdog timeout (limit 3min)!
Jul 16 17:14:10 pall systemd[1]: systemd-resolved.service: Watchdog timeout (limit 3min)!
Jul 16 17:14:10 pall systemd[1]: systemd-journald.service: Watchdog timeout (limit 3min)!
Jul 16 19:54:49 pall systemd[3844]: Failed to start update-notifier-crash.service - Notification regarding a crash report.
Jul 17 07:28:00 pall kernel: watchdog: BUG: soft lockup - CPU#0 stuck for 13913s! [Privileged Cont:7469]
Jul 17 07:28:00 pall kernel: rcu: INFO: rcu_preempt detected stalls on CPUs/tasks:
Jul 17 07:28:00 pall kernel: rcu: 0-...! (1 GPs behind) idle=45c4/1/0x4000000000000000 softirq=297719/297720
Jul 17 07:28:00 pall kernel: rcu: (detected by 1, t=14927239 jiffies, g=506377, q=7 ncpus=2)
Jul 17 07:28:01 pall kernel: rcu: rcu_preempt kthread timer wakeup didn't happen for 14927236 jiffies! g506377 f0x0 RCU
Jul 17 07:28:01 pall kernel: rcu: Possible timer handling issue on cpu=1 timer-sofirq=380641
Jul 17 07:28:01 pall kernel: rcu: rcu_preempt kthread starved for 14927239 jiffies! g506377 f0x0 RCU
Jul 17 07:28:01 pall kernel: rcu: Unless rcu_preempt kthread gets sufficient CPU time, it will keep
Jul 17 07:28:01 pall kernel: rcu: RCU grace-period kthread stack dump:
Jul 17 07:28:01 pall kernel: rcu: Stack dump where RCU GP kthread last ran:
Jul 17 07:28:01 pall kernel: rcu: rcu_preempt kthread stack dump:
```

```

pall@pall:~$ sudo journalctl -f
Oct 02 18:56:59 pall sudo[8834]: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000)
Oct 02 18:59:56 pall sudo[8834]: pam_unix(sudo:session): session closed for user root
Oct 02 18:59:58 pall sudo[8843]:    pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/journalctl -p err
Oct 02 18:59:58 pall sudo[8843]: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000)
Oct 02 19:00:00 pall systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
Oct 02 19:00:00 pall systemd[1]: sysstat-collect.service: Deactivated successfully.
Oct 02 19:00:00 pall systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
Oct 02 19:00:42 pall sudo[8843]: pam_unix(sudo:session): session closed for user root
Oct 02 19:00:49 pall sudo[8852]:    pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/journalctl -f
Oct 02 19:00:49 pall sudo[8852]: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000)

```

Screenshot 6: Authentication log evidence

```

2026-09-25T00:05:01.800652+01:00 pall CRON[39085]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)
2026-09-25T00:05:01.821191+01:00 pall CRON[39085]: pam_unix(cron:session): session closed for user root
2026-09-25T00:15:01.923828+01:00 pall CRON[39140]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)
2026-09-25T00:15:01.993705+01:00 pall CRON[39140]: pam_unix(cron:session): session closed for user root
2026-09-25T00:17:02.037378+01:00 pall CRON[39156]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)
2026-09-25T00:17:02.085312+01:00 pall CRON[39156]: pam_unix(cron:session): session closed for user root
2026-09-25T06:15:05.282106+01:00 pall CRON[39169]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)
2026-09-25T06:15:11.395698+01:00 pall CRON[39169]: pam_unix(cron:session): session closed for user root
2026-09-25T08:53:54.869151+01:00 pall CRON[39400]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)
2026-09-25T08:53:54.918032+01:00 pall CRON[39400]: pam_unix(cron:session): session closed for user root
2026-09-25T08:53:55.001487+01:00 pall CRON[39406]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)
2026-09-25T08:53:55.056583+01:00 pall CRON[39406]: pam_unix(cron:session): session closed for user root
2026-09-25T08:53:55.087075+01:00 pall CRON[39419]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)
2026-09-25T08:53:55.093650+01:00 pall CRON[39419]: pam_unix(cron:session): session closed for user root
2026-09-25T08:55:01.158711+01:00 pall CRON[39436]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)
2026-09-25T08:55:01.179343+01:00 pall CRON[39436]: pam_unix(cron:session): session closed for user root
2026-09-25T09:05:01.643184+01:00 pall CRON[46104]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)
2026-09-25T09:05:01.692719+01:00 pall CRON[46104]: pam_unix(cron:session): session closed for user root
2026-09-25T09:15:01.666239+01:00 pall CRON[60464]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)
2026-09-25T09:15:01.682099+01:00 pall CRON[60464]: pam_unix(cron:session): session closed for user root
2026-09-25T09:17:01.928205+01:00 pall CRON[60490]: pam_unix(cron:session): session opened for user root(uid=0) by root(u
id=0)

```

```

pall@pall:~$ sudo grep -i "failed password" /var/log/auth.log
2026-10-02T19:04:00.254990+01:00 pall sudo:    pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/grep -i
'failed password' /var/log/auth.log
pall@pall:~$

```

```

pall@pall:~$ sudo grep -i "sudo" /var/log/auth.log
2026-10-02T17:56:08.621057+01:00 pall sudo:      pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/systemc
tl status auditd
2026-10-02T17:56:08.639661+01:00 pall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000
)
2026-10-02T17:56:09.062839+01:00 pall sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T18:12:39.166171+01:00 pall sudo:      pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/apt up
ate
2026-10-02T18:12:39.194459+01:00 pall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000
)
2026-10-02T18:13:38.541008+01:00 pall sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T18:13:38.583442+01:00 pall sudo:      pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/apt in
tall auditd audispd-plugins lynis -y
2026-10-02T18:13:38.586329+01:00 pall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000
)
2026-10-02T18:14:08.537142+01:00 pall sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T18:15:13.538076+01:00 pall sudo:      pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/systemc
tl enable auditd
2026-10-02T18:15:13.542019+01:00 pall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000
)
2026-10-02T18:15:13.560635+01:00 pall sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T18:15:13.590269+01:00 pall sudo:      pall : TTY=pts/0 ; PWD=/home/pall ; USER=
bin/systemc
tl start auditd
2026-10-02T18:15:13.593513+01:00 pall sudo: pam_unix(sudo:session): session opened for us
l(uid=1000
)

```

Screenshot 7: Privileged access review

```

2026-09-25T00:00:05.032135+01:00 pall rsyslogd: [origin software="rsyslogd" swVersion="8.2312.0" x-pid="902" x-info="htt
ps://www.rsyslog.com"] rsyslogd was HUPed
2026-09-25T00:00:05.374469+01:00 pall kernel: audit: type=1400 audit(1790290805.370:266): apparmor="DENIED" operation="o
pen" class="file" profile="snap.firmware-updater.firmware-notifier" name="/proc/sys/vm/max_map_count" pid=38881 comm="fi
rmware-notifi" requested_mask="r" denied_mask="r" fsuid=1000 ouid=0
2026-09-25T00:00:05.535086+01:00 pall systemd[3910]: snap.firmware-updater.firmware-notifier.service: Consumed 3.680s CP
U time.
2026-09-25T00:05:01.817573+01:00 pall CRON[39086]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2026-09-25T00:07:00.417117+01:00 pall systemd[1]: Starting sysstat-summary.service - Generate a daily summary of process
accounting...
2026-09-25T00:07:00.958074+01:00 pall systemd[1]: sysstat-summary.service: Deactivated successfully.
2026-09-25T00:07:00.960326+01:00 pall systemd[1]: Finished sysstat-summary.service - Generate a daily summary of process
accounting.
2026-09-25T00:10:00.401358+01:00 pall systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-09-25T00:10:00.557373+01:00 pall systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-09-25T00:10:00.559525+01:00 pall systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2026-09-25T00:13:58.456448+01:00 pall kernel: workqueue: ata_sff_pio_task hogged CPU for >10000us 19 times, consider swi
tching to WQ_UNBOUND
2026-09-25T00:15:01.944821+01:00 pall CRON[39141]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2026-09-25T00:17:02.046562+01:00 pall CRON[39157]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
2026-09-25T00:20:02.875081+01:00 pall rtkit-daemon[1583]: The canary thread is apparently starving. Taking action.
2026-09-25T00:20:03.112371+01:00 pall rtkit-daemon[1583]: Demoting known real-time threads.
2026-09-25T00:20:03.147267+01:00 pall rtkit-daemon[1583]: Successfully demoted thread 33141 of process 32913.
2026-09-25T00:20:03.164547+01:00 pall rtkit-daemon[1583]: Successfully demoted thread 6439 of process 6429.
2026-09-25T00:20:03.176197+01:00 pall rtkit-daemon[1583]: Successfully demoted thread 6429 of process 6429.
2026-09-25T00:20:03.189749+01:00 pall rtkit-daemon[1583]: Successfully demoted thread 6431 of process 6428.
2026-09-25T00:20:03.211459+01:00 pall rtkit-daemon[1583]: Successfully demoted thread 642
2026-09-25T00:20:03.211992+01:00 pall rtkit-daemon[1583]: Successfully demoted thread 419
2026-09-25T00:20:03.284323+01:00 pall rtkit-daemon[1583]: Successfully demoted thread 395
2026-09-25T00:20:03.476530+01:00 pall rtkit-daemon[1583]: Successfully demoted thread 393
2026-09-25T00:20:03.477155+01:00 pall rtkit-daemon[1583]: Demoted 8 threads.

```

```

pall@pall:~$ sudo grep -i "sudo" /var/log/auth.log
2026-10-02T17:56:08.621057+01:00 pall sudo:      pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/systemc
tl status auditd
2026-10-02T17:56:08.639661+01:00 pall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000
)
2026-10-02T17:56:09.062839+01:00 pall sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T18:12:39.166171+01:00 pall sudo:      pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/apt up
date
2026-10-02T18:12:39.194459+01:00 pall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000
)
2026-10-02T18:13:38.541008+01:00 pall sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T18:13:38.583442+01:00 pall sudo:      pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/apt ins
tall auditd audispd-plugins lynis -y
2026-10-02T18:13:38.586329+01:00 pall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000
)
2026-10-02T18:14:08.537142+01:00 pall sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T18:15:13.538076+01:00 pall sudo:      pall : TTY=pts/0 ; PWD=/home/pall ; USER=root ; COMMAND=/usr/bin/systemc
tl enable auditd
2026-10-02T18:15:13.542019+01:00 pall sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pall(uid=1000
)
2026-10-02T18:15:13.560635+01:00 pall sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T18:15:13.590269+01:00 pall sudo:      pall : TTY=pts/0 ; PWD=/home/pall ; USER=
bin/systemc
tl start auditd
2026-10-02T18:15:13.593513+01:00 pall sudo: pam_unix(sudo:session): session opened for us
l(uid=1000
)
2026-10-02T18:15:13.605940+01:00 pall sudo: pam_unix(sudo:session): session closed for us

```

Screenshot 8: Error/warning review

```
pall@pall:~$ sudo grep -i "warning" /var/log/syslog
2026-09-25T09:20:29.621396+01:00 pall org.gnome.Nautilus[64445]: libEGL warning: failed to get driver name for fd -1
2026-09-25T09:20:29.623385+01:00 pall org.gnome.Nautilus[64445]: libEGL warning: MESA-LOADER: failed to retrieve device
information
2026-09-25T09:20:29.623499+01:00 pall org.gnome.Nautilus[64445]: libEGL warning: failed to get driver name for fd -1
2026-09-25T09:20:30.189373+01:00 pall org.gnome.Nautilus[64445]: libEGL warning: egl: failed to create dri2 screen
2026-10-02T17:29:47.725875+01:00 pall kernel: Speculative Return Stack Overflow: WARNING: See https://kernel.org/doc/htm
l/latest/admin-guide/hw-vuln/srso.html for mitigation options.
2026-10-02T17:29:47.732282+01:00 pall kernel: WARNING: kernel/rcu/tree_plugin.h:823 at rcu_sched_clock_irq+0x394/0x3a0,
CPU#0: (udev-worker)/393
2026-10-02T17:29:47.748919+01:00 pall apparmor.systemd[814]: Warning: found usr.sbin.sssd in /etc/apparmor.d/force-compl
ain, forcing complain mode
2026-10-02T17:29:47.748923+01:00 pall apparmor.systemd[814]: Warning from /etc/apparmor.d (/etc/apparmor.d/usr.sbin.sssd
line 69): Caching disabled for: 'usr.sbin.sssd' due to force complain
2026-10-02T17:30:06.670108+01:00 pall gnome-session-binary[1518]: gnome-session-binary[1518]: WARNING: Could not check if unit
gnome-session-wayland@gnome-login.target is active: Error calling StartServiceByName for org.freedesktop.systemd1: Proce
ss org.freedesktop.systemd1 exited with status 1
2026-10-02T17:30:06.670392+01:00 pall gnome-session-binary[1518]: WARNING: Could not check if unit gnome-session-wayland
@gnome-login.target is active: Error calling StartServiceByName for org.freedesktop.systemd1: Process org.freedesktop.sy
stemd1 exited with status 1
2026-10-02T17:29:43.009805+01:00 pall containerd[1337]: time="2026-10-02T17:29:43.009442724+01:00" level=info msg="loadi
ng plugin" id=io.containerd.warning.v1.deprecations type=io.containerd.warning.v1
2026-10-02T17:29:43.486838+01:00 pall snapd[904]: autorepair.go:177: WARNING: the system booted with an old initrd witho
ut unlocked status reporting
2026-10-02T17:29:43.805554+01:00 pall containerd[1337]: time="2026-10-02T17:29:43.8043789" level=info msg="start
ing cri plugin" config="{\"containerd\":{\"defaultRuntimeName\":\"runc\",\"runtimes\":{\"\"io.containerd.runc.v2\":{
\"runtimePath\":\"/usr/bin/runc\",\"PodAnnotations\":{\"\"ContainerAnnotations\":{\"\"io.containerd.runc.v2\":{\"\"CriuImagePath\":
\"/usr/bin/criu\",\"CriuWorkPath\":\"/usr/bin/criu\",\"IoGid\":0,\"IoUid\":0,\"NoNewKeyrings\":true,\"ShimC
```

Evidence Bundle 3 – Lynis Security Assessment

Objective

Evaluate the security posture and hardening status of the Linux system.

Command Executed

```
sudo lynis audit system
```

Summary of Results

Hardening Index: 60/100

```
Lynis security scan details:

Hardening index : 60 [#####          ]
Tests performed : 258
Plugins enabled : 1

Components:
- Firewall           [V]
- Malware scanner    [X]

Scan mode:
Normal [V]  Forensics [ ]  Integration [ ]  Pentest [ ]

Lynis modules:
- Compliance status  [?]
- Security audit     [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data                : /var/log/lynis-report.dat

=====

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)
```

Key Findings

1. The system achieved a Lynis Hardening Index score of 60/100, indicating that additional security hardening measures are required to improve the overall security posture.
2. Audit monitoring was configured; however, validation of loaded audit rules was restricted by environment limitations, resulting in reduced visibility into audit-event monitoring.

3. Authentication and system log reviews identified security-relevant events requiring continuous monitoring, including failed authentication attempts, privileged activity, and system warnings.

Prioritised Recommendations

Priority	Recommendation
High	Address identified security weaknesses
Medium	Improve configuration hardening settings
Low	Review and optimize system configuration

Insert Screenshots

Screenshot 9: Lynis scan execution

```
pall@pall:~$ sudo lynis audit system

[ Lynis 3.0.9 ]

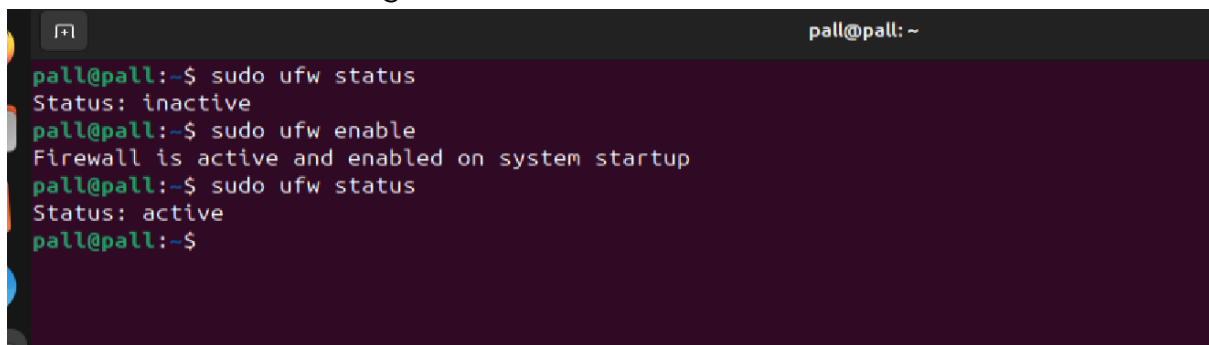
#####
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
#####

[+] Initializing program
-----
- Detecting OS... [ DONE ]
- Checking profiles... [ DONE ]

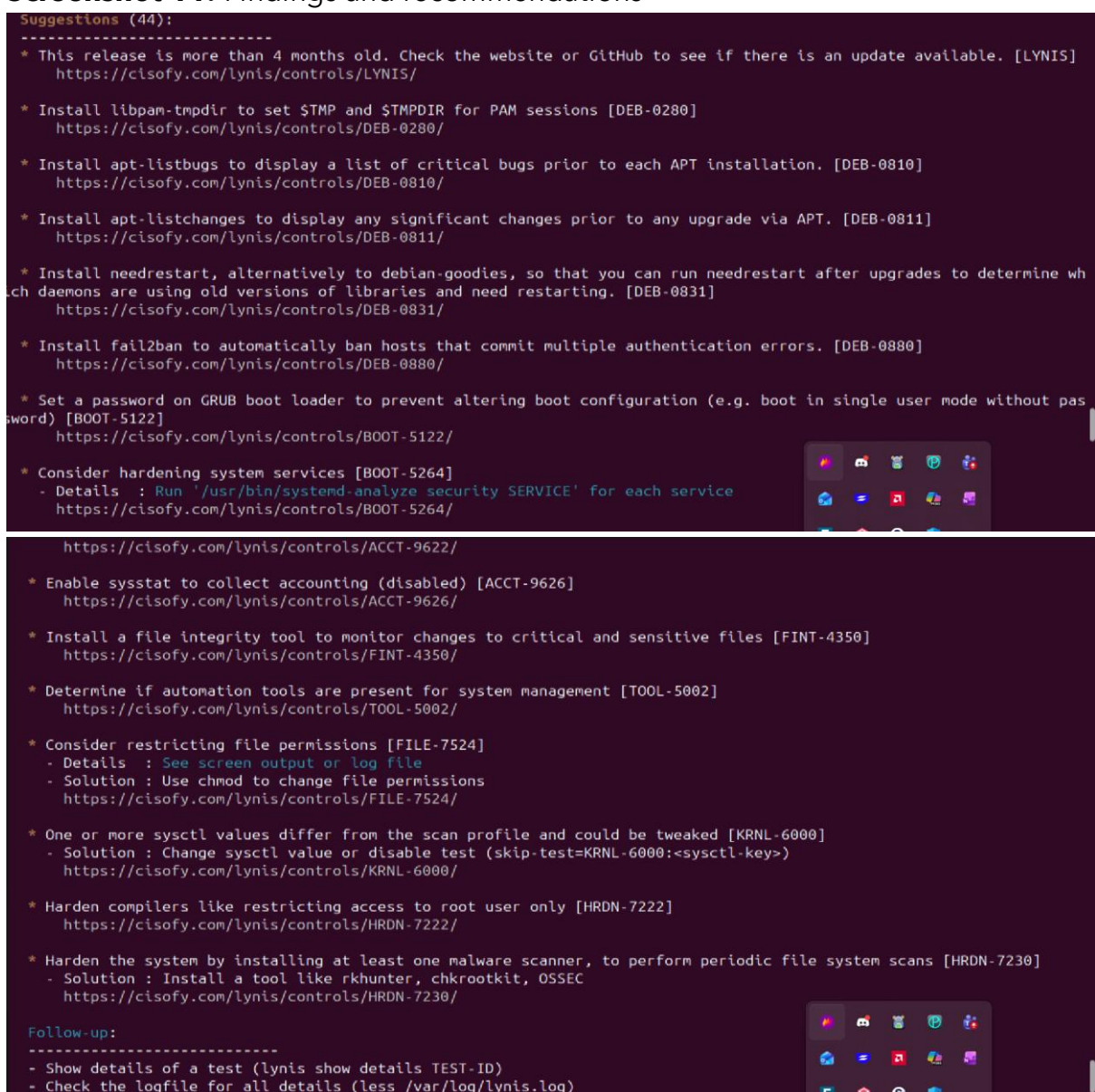
-----
Program version:      3.0.9
Operating system:     Linux
Operating system name: Ubuntu
Operating system version: 24.04
Kernel version:       7.0.0
Hardware platform:    x86_64
Hostname:             pall
-----
Profiles:             /etc/lynis/default.prf
Log file:              /var/log/lynis.log
Report file:          /var/log/lynis-report.dat
```

Screenshot 10: Hardening Index



```
pall@pall:~  
pall@pall:~$ sudo ufw status  
Status: inactive  
pall@pall:~$ sudo ufw enable  
Firewall is active and enabled on system startup  
pall@pall:~$ sudo ufw status  
Status: active  
pall@pall:~$
```

Screenshot 11: Findings and recommendations



```
Suggestions (44):  
-----  
* This release is more than 4 months old. Check the website or GitHub to see if there is an update available. [LYNIS]  
  https://cisofy.com/lynis/controls/LYNIS/  
  
* Install libpam-tmpdir to set $TMP and $TMPDIR for PAM sessions [DEB-0280]  
  https://cisofy.com/lynis/controls/DEB-0280/  
  
* Install apt-listbugs to display a list of critical bugs prior to each APT installation. [DEB-0810]  
  https://cisofy.com/lynis/controls/DEB-0810/  
  
* Install apt-listchanges to display any significant changes prior to any upgrade via APT. [DEB-0811]  
  https://cisofy.com/lynis/controls/DEB-0811/  
  
* Install needrestart, alternatively to debian-goodies, so that you can run needrestart after upgrades to determine wh  
ch daemons are using old versions of libraries and need restarting. [DEB-0831]  
  https://cisofy.com/lynis/controls/DEB-0831/  
  
* Install fail2ban to automatically ban hosts that commit multiple authentication errors. [DEB-0880]  
  https://cisofy.com/lynis/controls/DEB-0880/  
  
* Set a password on GRUB boot loader to prevent altering boot configuration (e.g. boot in single user mode without pas  
sword) [BOOT-5122]  
  https://cisofy.com/lynis/controls/BOOT-5122/  
  
* Consider hardening system services [BOOT-5264]  
  - Details : Run '/usr/bin/systemd-analyze security SERVICE' for each service  
  https://cisofy.com/lynis/controls/BOOT-5264/  
  
https://cisofy.com/lynis/controls/ACCT-9622/  
  
* Enable sysstat to collect accounting (disabled) [ACCT-9626]  
  https://cisofy.com/lynis/controls/ACCT-9626/  
  
* Install a file integrity tool to monitor changes to critical and sensitive files [FINT-4350]  
  https://cisofy.com/lynis/controls/FINT-4350/  
  
* Determine if automation tools are present for system management [TOOL-5002]  
  https://cisofy.com/lynis/controls/TOOL-5002/  
  
* Consider restricting file permissions [FILE-7524]  
  - Details : See screen output or log file  
  - Solution : Use chmod to change file permissions  
  https://cisofy.com/lynis/controls/FILE-7524/  
  
* One or more sysctl values differ from the scan profile and could be tweaked [KRNL-6000]  
  - Solution : Change sysctl value or disable test (skip-test=KRNL-6000:<sysctl-key>)  
  https://cisofy.com/lynis/controls/KRNL-6000/  
  
* Harden compilers like restricting access to root user only [HRDN-7222]  
  https://cisofy.com/lynis/controls/HRDN-7222/  
  
* Harden the system by installing at least one malware scanner, to perform periodic file system scans [HRDN-7230]  
  - Solution : Install a tool like rkhunter, chkrootkit, OSSEC  
  https://cisofy.com/lynis/controls/HRDN-7230/  
  
Follow-up:  
-----  
- Show details of a test (lynis show details TEST-ID)  
- Check the logfile for all details (less /var/log/lynis.log)
```

Interpretation

The system demonstrated a reasonable security posture but would benefit from implementation of the identified hardening recommendations.

Evidence Bundle 4 – Control Monitoring and Governance

Control Monitoring Table

Control / Objective	Evidence	Owner	Status	KPI/KRI	Risk / Significance	Remediation	Retest
Authentication Monitoring	auth.log failed logins	Security Operations	Needs Improvement	>5 failed logins/day	Possible brute-force attacks	Investigate failed attempts	Weekly
Privileged Access Monitoring	sudo entries	Linux Administrator	Effective	All privileged activity logged	Unauthorized privilege abuse	Review access rights	Weekly
Audit Coverage	auditctl output	Linux Administrator	Partially Effective	Audit rules successfully loaded	Reduced visibility due to audit restrictions	Validate audit capability	Monthly
Security Hardening	Lynis findings	Security Team	Improvement Required	Hardening Index >80	Increased attack surface	Implement recommendations	Monthly
Log Monitoring	journalctl review	IT Operations	Effective	Critical logs reviewed daily	Delayed incident detection	Continuous monitoring	Daily

Governance Escalation Questions

1. Which finding is highest priority?

Authentication failures are the highest priority because they may indicate attempted unauthorized access or credential attacks.

2. Who should own remediation?

Finding	Owner
Authentication failures	Security Operations
Audit visibility gaps	Linux Administrator
Hardening weaknesses	Security Team

3. Escalation Triggers

Condition	Escalation
More than 10 failed logins within 24 hours	SOC Manager
Audit logging disabled	CISO
Critical Lynis finding	CISO and Risk Function
Unauthorized privileged activity	Management Committee

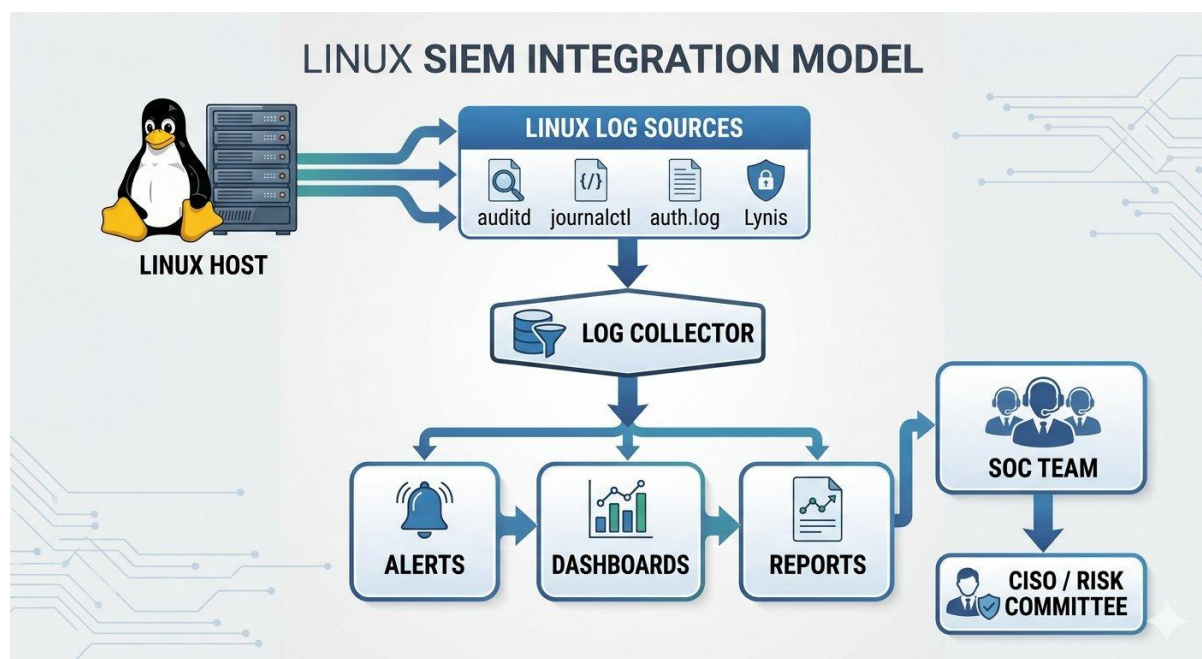
4. Evidence of Successful Remediation

- Reduction of failed authentication attempts.
- Improved Lynis Hardening Index.
- Successful audit rule validation.
- Clean follow-up log review.

5. Retesting

Control	Method	Frequency
Authentication Monitoring	Log review	Weekly
Audit Controls	auditctl validation	Monthly
Hardening Controls	Lynis audit	Monthly
Logging Controls	journalctl review	Daily

Evidence Bundle 5 - SIEM and Automation Mapping



Automated Monitoring Examples

Event	Automated Action
Failed login attempts	Generate alert
Privileged account activity	Create SOC investigation
Critical Lynis finding	Escalate to Security Team
Audit logging failure	Escalate to CISO
Repeated security errors	Open incident ticket

Evidence Bundle 6 – Final Audit Findings

Key Findings

1. Authentication failures were identified and require monitoring.
2. Audit subsystem access was partially restricted by environment limitations.
3. Lynis identified hardening opportunities.

Recommendations

1. Implement continuous monitoring of authentication activity.

2. Validate audit controls in a fully supported Linux environment.
3. Implement prioritized Lynis recommendations.
4. Centralize Linux logs within a SIEM.
5. Establish routine security control retesting.

Remediation Plan

Finding	Owner	Target Action
Authentication failures	Security Operations	Monitor and investigate
Audit limitations	Linux Administrator	Improve audit visibility
Hardening findings	Security Team	Implement recommendations

Conclusion

The assessment demonstrated that Linux monitoring and auditing controls were functioning and capable of supporting security oversight activities. Log review, auditing configuration, and security assessment activities provided evidence supporting operational monitoring and governance processes. While improvements are required in audit visibility and system hardening, the overall control environment provides a foundation for continuous monitoring, security assurance, and governance reporting.

References

- Lynis Documentation. CISOfy.
- Ubuntu Linux Documentation.
- Linux Audit Framework (auditd) Documentation.